POWER PULSE

CRO Important Formulae

Notes by BALAJI SIR

1 Electron beam

1. $$v = \sqrt{\frac{2eV_a}{m}} \approx 5.93 \times 10^5 \sqrt{V_a} \ \text{m/s}$$

V_a = accelerating voltage

2. $$S = \frac{\ell L}{2\,d\,V_a} \ \text{(m/V)}, \qquad D = S\,V_d$$

electrostatic deflection sensitivity; ℓ = plate length, L = plate centre to screen, d = plate separation

2 Reading the screen

3. $T = (\text{divisions}) \times (\text{time/div}), \ f = \frac{1}{T}; \qquad V_{pp} = (\text{divisions}) \times (\text{volts/div}) \times \text{probe factor}$

4. $$V_{rms} = \frac{V_{pp}}{2\sqrt{2}} \ \text{(sine)}$$

3 Lissajous figures

5. $$\frac{f_y}{f_x} = \frac{\text{points touching a horizontal tangent}}{\text{points touching a vertical tangent}}$$

a straight line means equal frequencies; the figure is stationary only at exact ratios

6. $$\sin\phi = \frac{y_{int}}{y_{max}}, \qquad \phi = 0 : \text{line}; \ \ \phi = 90^\circ \ \text{(equal amplitudes)} : \text{circle}$$

phase difference from the ellipse; y_int = intercept on the y-axis

4 Bandwidth and rise time

7. $$t_r = \frac{0.35}{BW}, \qquad t_{r,\text{observed}} = \sqrt{t_{r,\text{signal}}^2 + t_{r,\text{scope}}^2}$$

the scope bandwidth should be about 5 times the signal frequency

5 Probe

8. $\times 10$ probe: series $9\,M\Omega$ with the $1\,M\Omega$ input; input C reduced 10 times

increases input impedance and reduces loading